AI-POWERED ORAL EXAMINATION PLATFORM

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Capstone Project

BACKGROUND & MOTIVATION

- Rapid growth of generative AI (e.g., ChatGPT) challenges traditional written assessments
- Increased risk of academic dishonesty in take-home exams
- Oral exams are more reliable but:
  - Time-consuming
  - Not scalable
- Need for a scalable, automated, and fair assessment solution

Goal: Reimagine oral exams using AI

PROBLEM STATEMENT

- Written exams no longer reliably measure true understanding
- Manual oral exams:
  - Require significant instructor time
  - Lack scalability for large classes
- No existing system combines:
  - Voice interaction
  - Automated evaluation
  - Real-time feedback

SOLUTION OVERVIEW

Designed for scalability + academic integrity

- Voice-based question delivery
- Student spoken response capture
- Automated evaluation using LLMs
- Real-time feedback generation

SYSTEM ARCHITECTURE

Frontend (React)

- Exam interface
- Audio recording

Backend (Python)

- API handling
- Data processing

Database (SQLite)

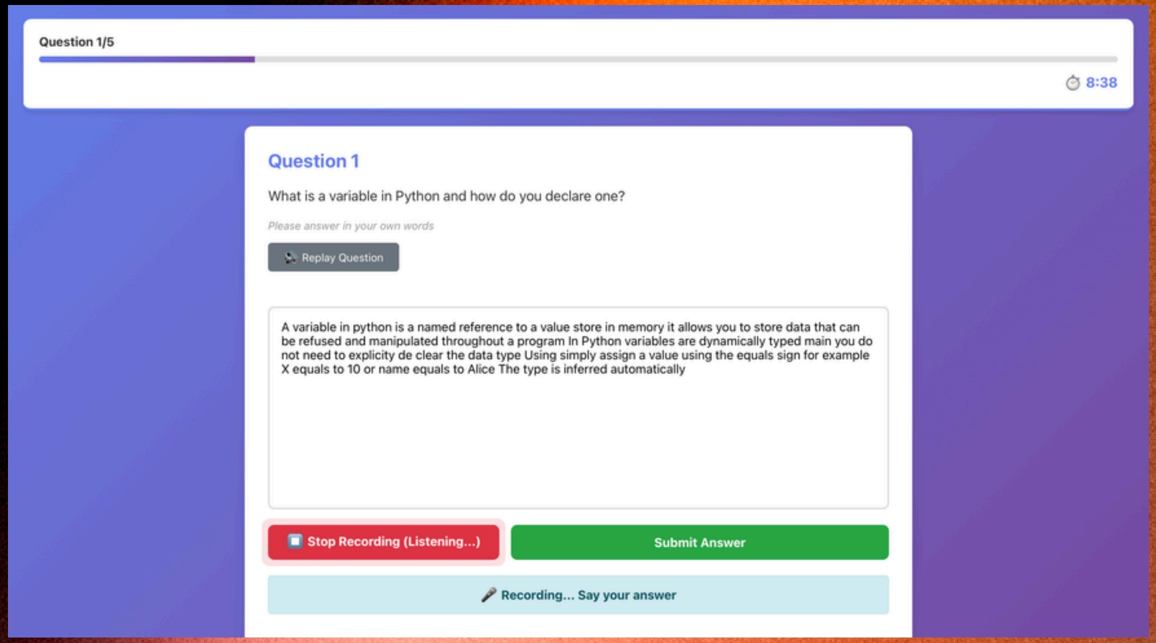
- User & exam records

AI Layer

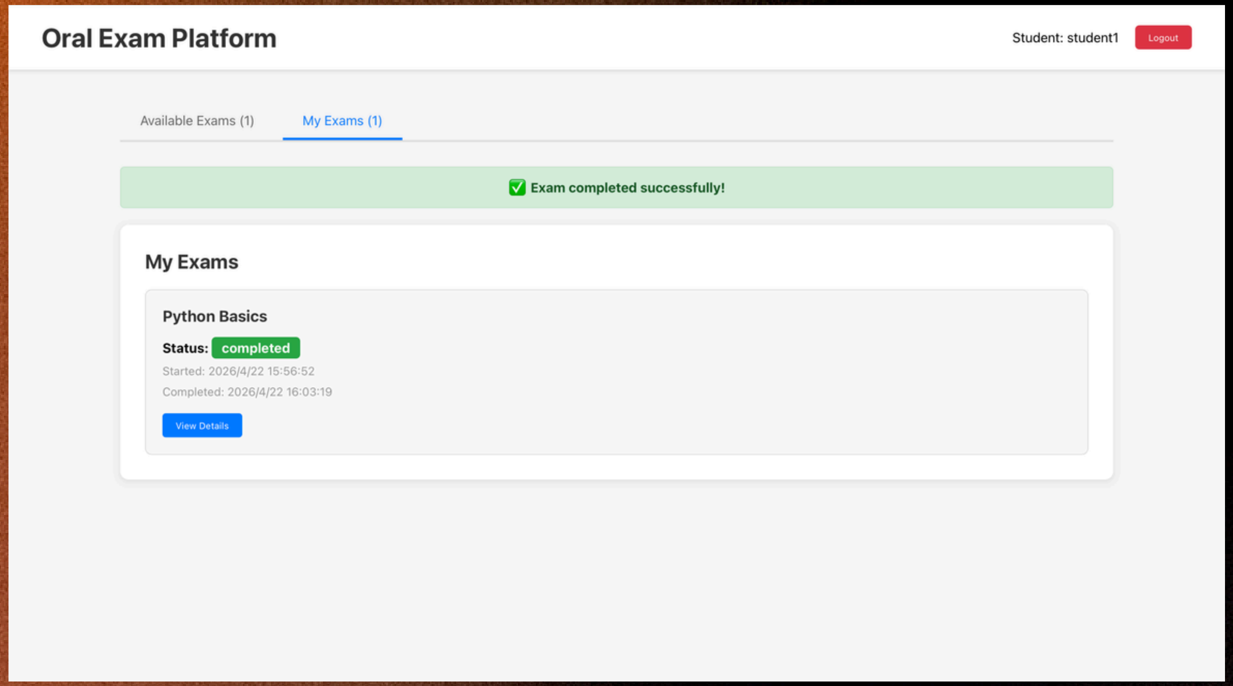
- LLM-based scoring (ChatGPT)

KEY FEATURES

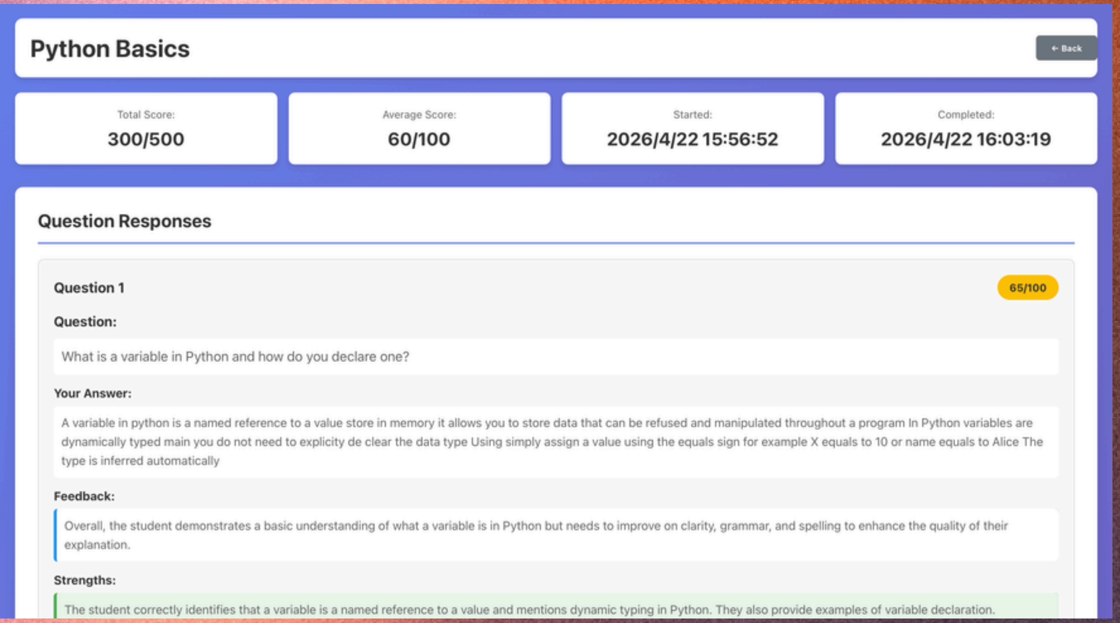
Automated oral question delivery



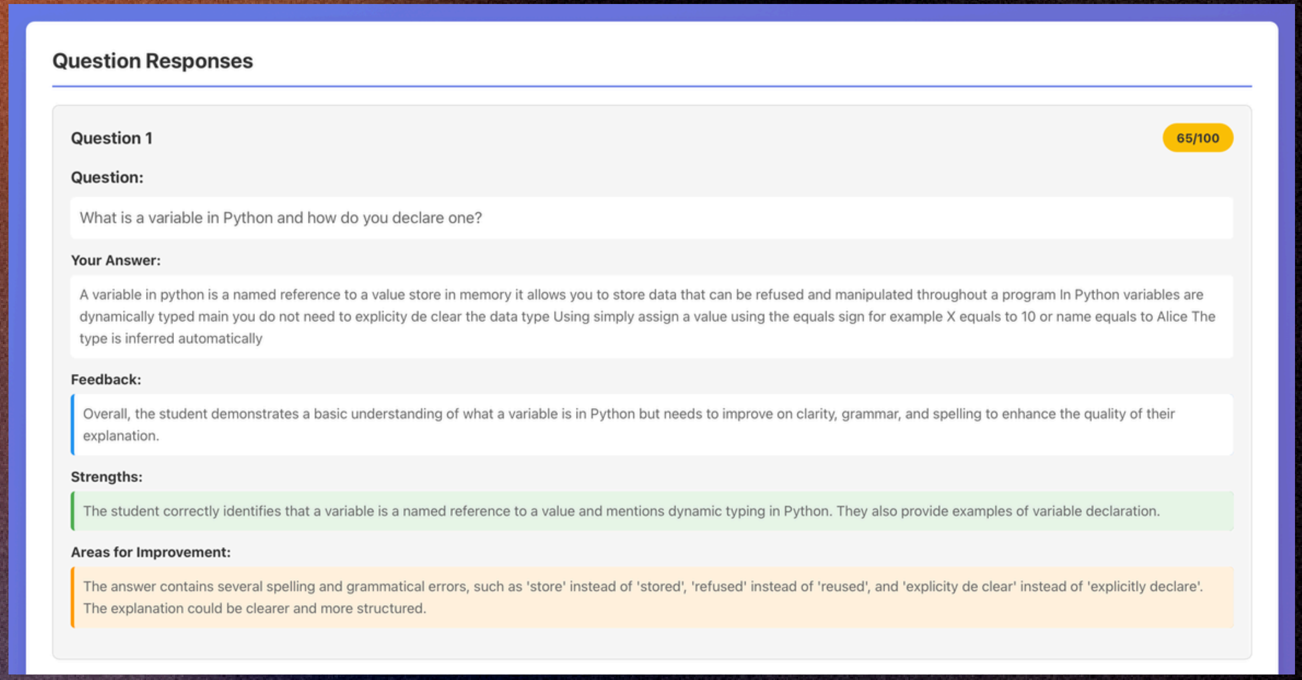
Real-time voice interaction



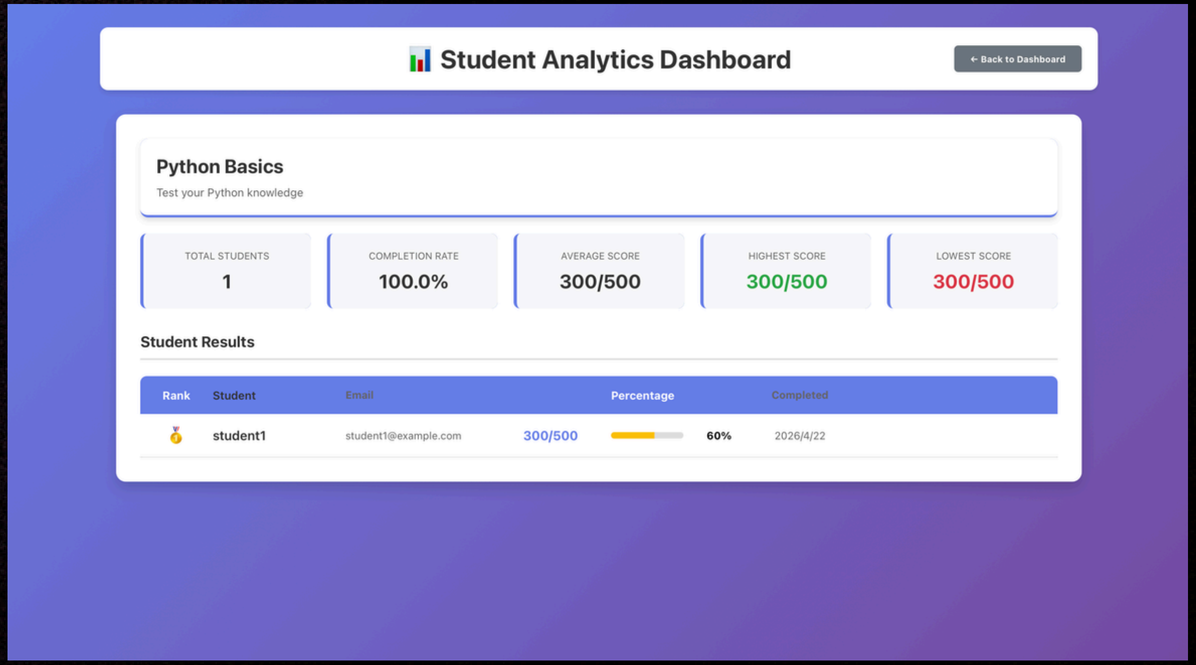
Multi-agent evaluation (LLM ensemble)



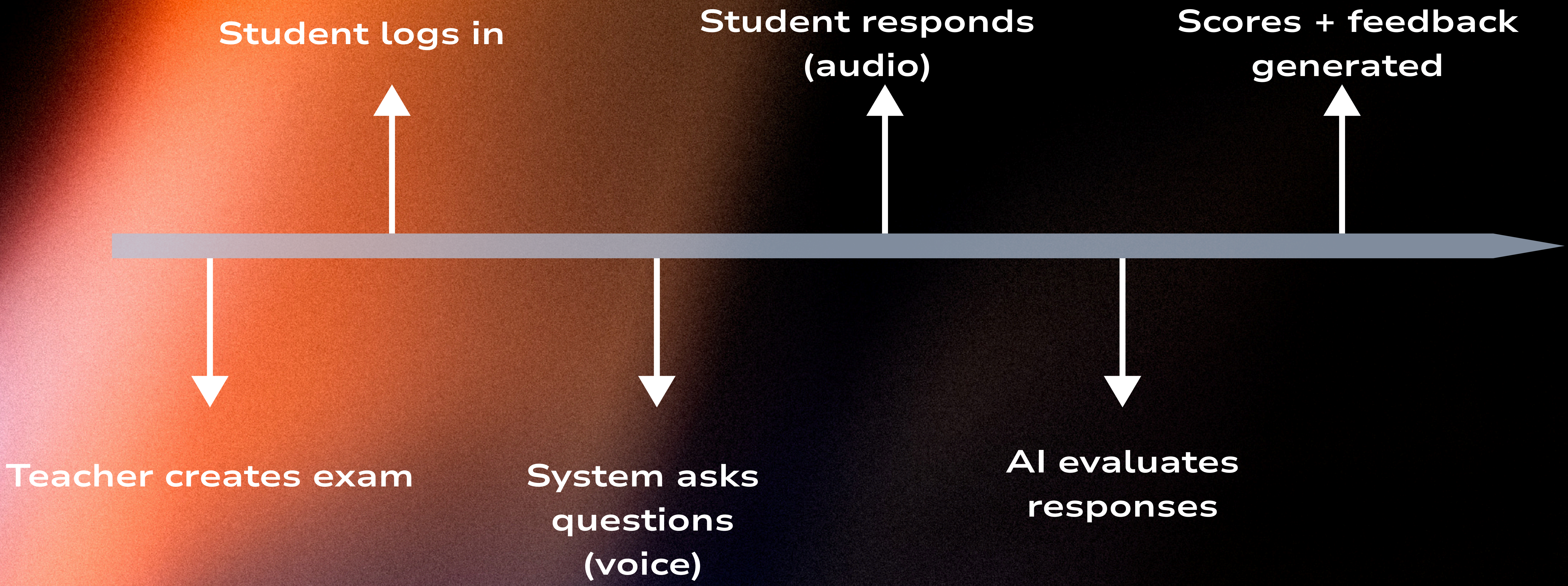
Personalized feedback



Teacher analytics dashboard



WORKFLOW



End-to-end automated oral exam

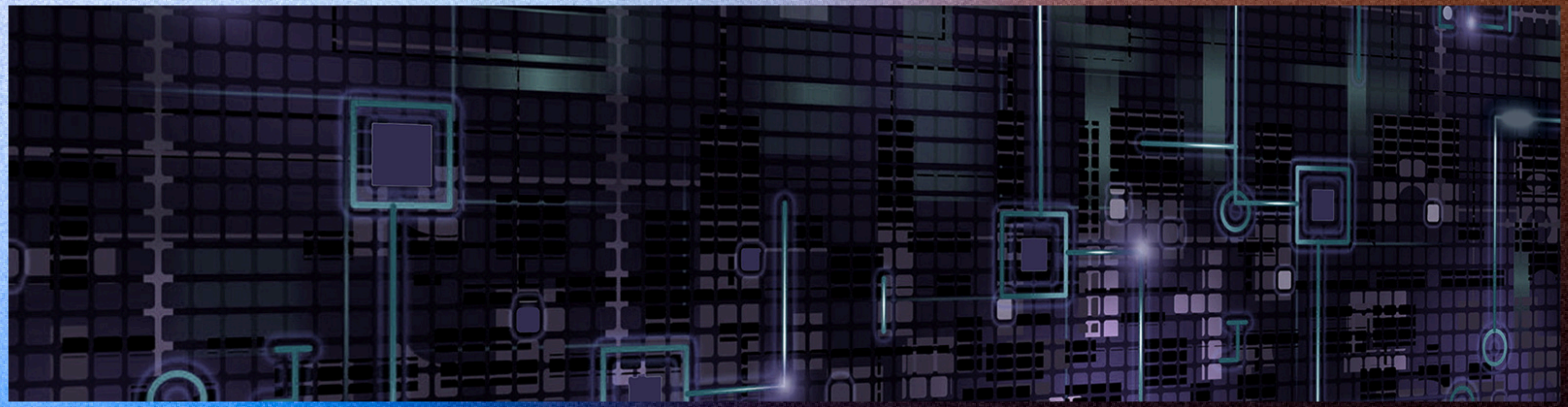
METHODOLOGY

- Built full-stack prototype
- Designed simulated dataset:
  - 5 students × 5 questions
- Defined answer quality levels:
  - High / Medium / Low
- Conducted multiple evaluation runs
  Tested consistency and scoring reliability

	A	B	C	D	E	F	G	H
1	Student	Question 1	Question 2	Question 3	Question 4	Question 5	Avg Score	Std Dev
2	Student 1 (High)	95	90	90	90	90	91	2.23606798
3	Student 2 (Medium)	90	85	80	80	80	83	4.47213595
4	Student 3 (Low)	60	60	60	60	60	60	0
5	Student 4 (Mixed: 2H/2M/1L)	95	90	80	80	60	81	13.4164079
6	Student 5 (Mixed: 1H/2M/2L)	90	85	80	60	60	75	14.1421356

	A	B	C	D	E
1	Response Type	Question	Attempt 1	Attempt 2	Attempt 3
2	High	Q1	95	95	90
3	Medium	Q3	80	80	80
4	Low	Q5	60	60	60

EVALUATION METRICS & RESULTS



Evaluation Metrics

- **Scoring Consistency**

→ Stability of AI evaluation across repeated runs

- **Evaluation Accuracy**

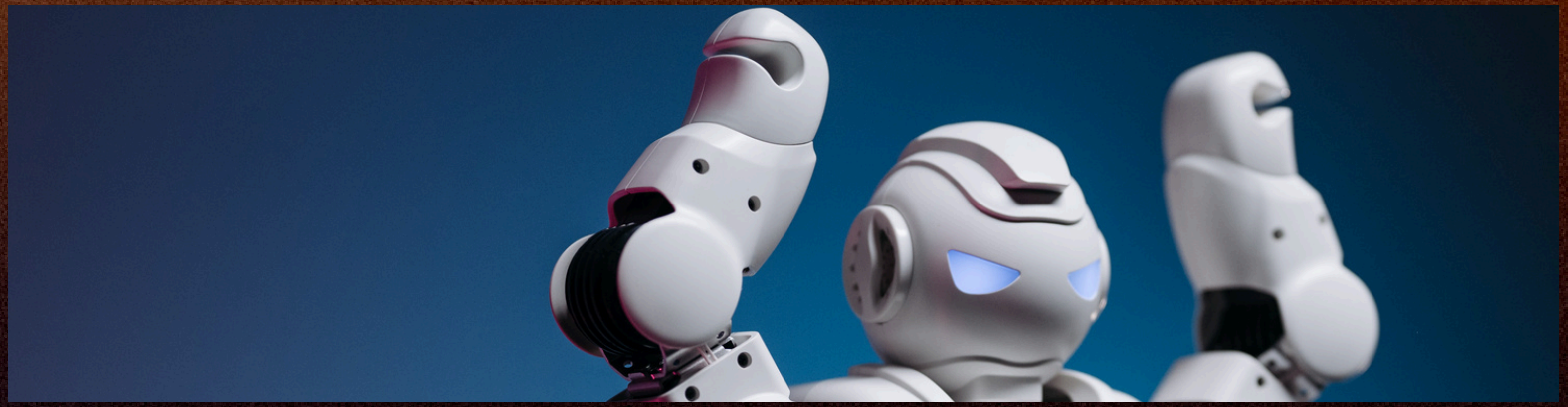
→ Ability to distinguish High / Medium / Low quality responses

- **Variability (Std. Dev.)**

→ Measure of scoring dispersion

- **System Usability**

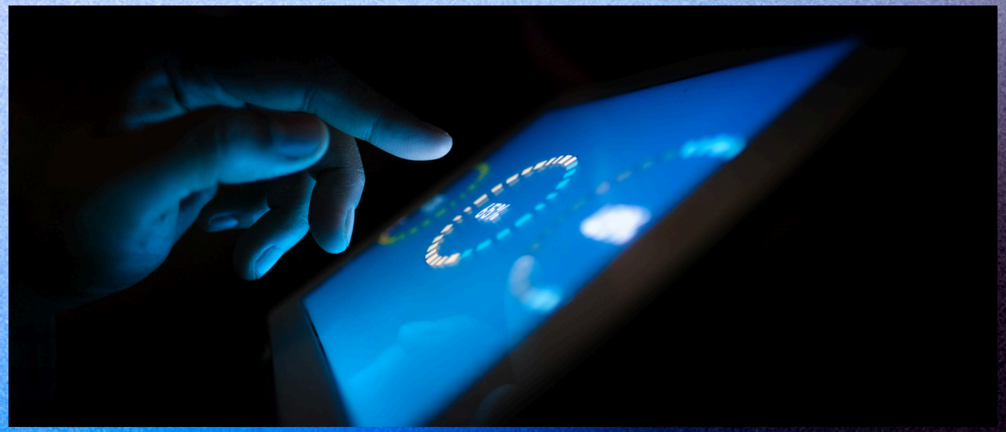
→ Ease of use for both students and instructors



Results

- AI successfully differentiated High, Medium, and Low responses
- Average simulated scores improved (~60 → ~80)
- Repeated evaluations showed relatively stable scoring patterns
- Low standard deviation indicated acceptable consistency
- End-to-end workflow validated system feasibility

CHALLENGES, LESSONS LEARNED & LIMITATIONS

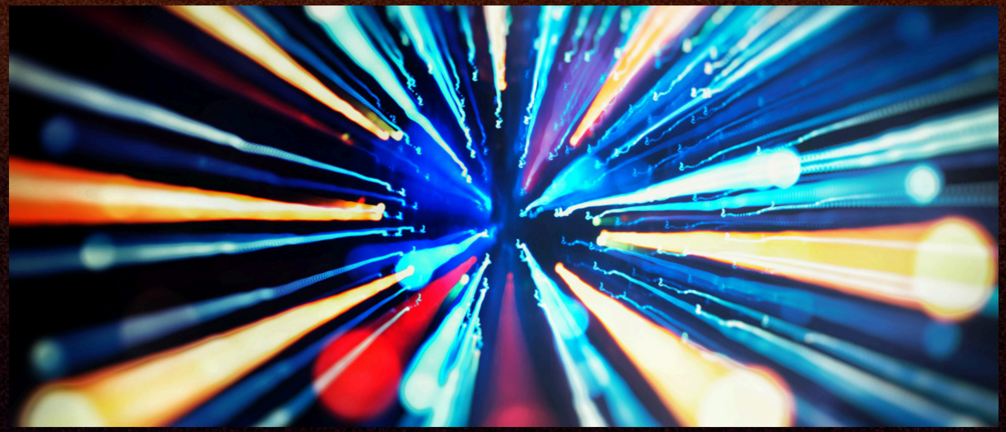


Challenges

- Variability in LLM-generated evaluations across runs
- Ensuring fairness and consistency in scoring
- Handling audio recording and processing reliability
- Integration complexity across frontend, backend, and AI models

Lessons Learned

- Prompt design significantly impacts evaluation quality
- Multi-model (LLM ensemble) improves robustness
- Voice interaction enhances authenticity of assessment
- Modular system design supports scalability and flexibility



Limitations

- Limited testing with simulated data rather than real students
- Dependence on LLM performance and potential bias
- Not fully optimized for production-level deployment
- Evaluation consistency still requires further validation

CONCLUSION & FUTURE WORK

Conclusion

- Developed a scalable AI-powered oral examination platform
- Demonstrated feasibility of automated voice-based assessment
- Addressed academic integrity challenges in the generative AI era
- Showed potential for improving efficiency, scalability, and fairness in evaluation

Future Work

- Conduct testing with real student data at larger scale
- Improve evaluation consistency and bias mitigation
- Integrate with Learning Management Systems (e.g., Brightspace)
- Enhance speech recognition and response analysis accuracy
- Deploy as a production-ready, cloud-based platform

THANK YOU

GitHub Repository:
https://github.com/danxia2002/oral-exam-platform.git

Includes:
- Source code
- Final report
- Testing data
- Presentation slides